Luca Zanolini

Researcher & Engineer · Distributed Systems and AI Safety

Now applying fault-tolerance and adversarial-analysis methods to multi-agent AI and AI safety

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PROFILE

I am a distributed-systems researcher: I design fault-tolerant protocols, prove them correct, and test specifications against adversarial and network assumptions to find where guarantees break. At the Ethereum Foundation (2022–2026) I co-designed Byzantine fault-tolerant consensus protocols and led an adversarial security evaluation of a major protocol upgrade that uncovered specification bugs, since fixed upstream. I now apply the same discipline – explicit fault models, safety and liveness guarantees, adversarial testing – to multi-agent AI systems and AI safety, building research prototypes in Rust and Python with AI assistance. Trained as a mathematician; PhD in Computer Science from the University of Bern; 20+ peer-reviewed publications, including at ACM CCS, PODC, IEEE ICDCS, and IEEE CSF.

CURRENT RESEARCH (2026)

- **Correlated Failures of AI Approval Monitors** (preprint, 2026): models AI approval monitors as a fail-prone system; attacks spread only through shared evidence, and rules that count independent sources, not votes, block far fewer correct payments at equal safety.
- **Reliable Distributed Agent Systems** (in progress, 2026–): an open-source research prototype applying distributed-systems rigor to multi-agent AI infrastructure – a replicated coordination plane with sandboxed agent workers treated as untrusted clients.
- **Concurrent and Distributed Systems in Rust** (2026–): an independent open-source, 12-module course developing distributed-systems theory through working Rust implementations, from Raft to Byzantine broadcast, with scripted failure demonstrations.

EXPERIENCE

Research Scientist **2022-2026**

Ethereum Foundation, Protocol Consensus Seoul / Remote

- Led an adversarial security evaluation of EIP-7732 (ePBS), a major planned redesign of Ethereum's block production: scoped the programme, identified ten externally observable security properties, formalized the adversarial and network assumptions each requires, and checked every claim against the executable specification with open models and proofs. Found concrete specification bugs, fixed upstream via `consensus-specs` pull requests.
- Co-designed Byzantine fault-tolerant consensus protocols that remain safe and live as participants go offline and rejoin, including 3-Slot Finality, which reduces time to finality from 64–95 slots to three under favourable conditions.
- Co-authored the fast confirmation rule – when a client can safely treat a block as confirmed before finality – from analysis through specification and tests; analyzed failure modes arising from interactions among concurrent protocol upgrades.
- Made safety, liveness, latency, and accountability trade-offs explicit to guide protocol teams' design and roadmap decisions.
- Main technical contact for the EF Academic Grants Round (with Duke, Stanford, a16z crypto, IC3); co-organized research workshops; mentored interns and graduate researchers.

Visiting Scholar **Sep 2025-Aug 2026**

University of Cambridge, Department of Computer Science and Technology Cambridge, UK

- Weekly small-group supervision for Concurrent and Distributed Systems; invited seminar and guest teaching on blockchain and Ethereum.

Research and Teaching Assistant **2019-2023**

University of Bern, Cryptology and Data Security Group Bern, Switzerland

- Developed theory and algorithms for Byzantine fault tolerance when participants hold different trust assumptions (asymmetric trust): quorum systems, reliable broadcast, and consensus; ran tutorials, exercises, and assessment for Cryptography and for Privacy and Data Security.

Research Intern **Summer 2021**

Chainlink Remote

- Built Go research prototypes and participated in distributed-systems and blockchain research seminars.

Ethereum Developer **2017-2019**

Interlogica – Solidity smart contracts in production Italy

EDUCATION

PhD in Computer Science, cum laude **2019-2023**

University of Bern – Advisor: Prof. Christian Cachin Bern, Switzerland

MSc in Mathematics, 110/110 cum laude **2015-2017**

Catholic University of the Sacred Heart Italy

BSc in Mathematics, 110/110 **2012-2015**

Catholic University of the Sacred Heart Italy

SELECTED PUBLICATIONS

Nine of 20+ peer-reviewed papers, plus a recent preprint; full list on DBLP and Google Scholar.

Correlated Failures of AI Approval Monitors
Luca Zanolini. arXiv preprint 2610.29178, 2026. arXiv

Accountable Liveness
Andrew Lewis-Pye, Joachim Neu, Tim Roughgarden, **Luca Zanolini**. ACM CCS 2025. doi

DAG-based Consensus with Asymmetric Trust
Ignacio Amores-Sesar, Christian Cachin, Juan Villacis, **Luca Zanolini**. ACM PODC 2025. doi

TOB-SVD: Total-Order Broadcast with Single-Vote Decisions in the Sleepy Model
Francesco D'Amato, Roberto Saltini, Thanh-Hai Tran, **Luca Zanolini**. IEEE ICDCS 2025. doi

Majorum: Ebb-and-Flow Consensus with Dynamic Quorums
Francesco D'Amato, Roberto Saltini, Thanh-Hai Tran, Yann Vonlanthen, **Luca Zanolini**. Financial Cryptography 2026, forthcoming. arXiv

Low-Latency Dynamically Available Total Order Broadcast
Sravya Yandamuri, Nibesh Shrestha, **Luca Zanolini**, Kartik Nayak. Financial Cryptography 2026, forthcoming. eprint

Data Independent Order Policy Enforcement: Limitations and Solutions
Sarisht Wadhwa, **Luca Zanolini**, Aditya Asgaonkar, Francesco D'Amato, Chengrui Fang, Fan Zhang, Kartik Nayak. ACM CCS 2024. doi

Asynchrony-Resilient Sleepy Total-Order Broadcast Protocols
Francesco D'Amato, Giuliano Losa, **Luca Zanolini**. ACM PODC 2024. doi

Recent Latest Message Driven GHOST: Balancing Dynamic Availability With Asynchrony Resilience
Francesco D'Amato, **Luca Zanolini**. IEEE CSF 2024. doi

Asymmetric Distributed Trust
Orestis Alpos, Christian Cachin, Björn Tackmann, **Luca Zanolini**. *Distributed Computing*, 37(3):247–277, 2024. doi

SELECTED TALKS AND WRITING

What Does ePBS Actually Guarantee? — **Aug 2026**
KAIST School of Computing – Invited research seminar — Daejeon, Korea

The Whys Behind Ethereum's Consensus Design — **Apr 2026**
BUIDL Asia 2026 — Seoul, Korea

Towards a Faster Finality Protocol for Ethereum — **Mar 2025**
PaPoC 2025 – Keynote — Rotterdam, Netherlands

A Fast Confirmation Rule for the Ethereum Consensus Protocol — **2024**
Devcon — Bangkok, Thailand

The Ethereum Consensus Protocol — **2023**
Yale University – Invited talk — New Haven, USA

Recent Latest Message Driven GHOST — **2023**
a16z Crypto Research Summer School — New York, USA

- Writing: Finality Is in the Eye of the Beholder (2025, with Vitalik Buterin and Ittai Abraham); Why Ethereum Needs a Dynamically Available Protocol (2025); research notes on 3-Slot Finality, fast confirmation, and upgrade interactions (2023–2025).

MENTORING AND COMMUNITY

- Supervised and mentored Ethereum Foundation research interns and an industrial PhD intern; mentored master's students; weekly small-group supervision of University of Cambridge students in Concurrent and Distributed Systems (2025–2026).
- Organized PhD-group research seminars at the University of Bern; co-organized Ethereum Foundation research workshops and cross-institutional protocol discussions.

ACADEMIC SERVICE

Program committees and reviewing: DISC (2022, 2025); OPODIS (2024, 2025); PaPoC (2025); Science of Blockchain Conference (2026); IEEE SRDS (2026); Financial Cryptography (2026); Asiacrypt (2020); *Distributed Computing* (2021).

SKILLS

Methods: formal and adversarial analysis of protocols | fault models | safety and liveness proofs | Byzantine fault tolerance | consensus and replication | concurrency | cryptography | specification testing

Prototyping: Rust | Go | Python | AI-assisted prototyping and specification testing (Claude Code, OpenAI Codex) | self-hosted agent systems (Hermes Agent: a team of eleven agents with scoped tools, scheduled jobs, and a self-built Matrix chat front end)

Collaboration: cross-team research across geographies and time zones